Tension-Based Spatial Localization for Long-Duration Elevated Payload Platforms

Experimental Report

Author: Suyash Prakash, Andy Ruina **Institution:** Plaksha University

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Abstract

This experiment investigates whether a point in three-dimensional space can be spatially constrained using a tensioned tether system without requiring GPS-based position control of the aerial platform. A 5-inch LW quadrotor equipped with GPS and an INAV flight controller was used as the aerial component. Three fixed ground anchors were connected to a common central junction, with a fourth tether connecting the junction to the drone.

The drone was operated in Angle mode with throttle above the free-hover requirement, causing the upper tether to become taut. After establishing tension, the controller input was left essentially unchanged. Although the drone moved or loitered around the region above the junction, the central junction remained constrained by the tether geometry. The main finding was that GPS position hold was not required to maintain the position of the central junction. Angular Control was more than enough to maintain the upper tether under tension, while the three ground tethers constrained the junction from below.

The result motivates a future blimp-kite platform designed to support an approximately 1 kg payload for long durations while minimizing energy consumption.

Introduction

Temporary elevated infrastructure can be useful for communication relays, RF systems, sensing equipment, and other electronic payloads. Conventional towers provide positional stability but are difficult to deploy, while multi-rotors are easy to deploy but consume significant energy continuously.

This project investigates an alternative in which the payload position is constrained primarily by a mechanical tether structure rather than active positional control of the aerial platform.

The proposed arrangement consists of three fixed ground anchors connected to a common central junction, with a fourth tether connecting this junction to an aerial platform. The junction is intended to act as the mounting point for a future elevated payload.

The initial experiment used a quadrotor to determine whether GPS-based position hold was actually necessary for maintaining this junction.

Experimental Setup

A 5-inch LW quadrotor equipped with GPS and an INAV flight controller was used. GPS lock and altitude-hold functionality were verified before tether testing.



Figure 1: 5-inch LW quadrotor used as the aerial platform.

Three ground anchor points were established using bricks fitted with eye bolts. Each anchor was connected by a string to a common central junction. A fourth string connected the junction to the drone.



Figure 2: Three-point ground anchoring arrangement used during testing.

Experimental Procedure

The drone was first flown normally to verify GPS lock and basic flight operation. For tether testing, the drone was placed in **Angle mode** and connected to the central junction.

Throttle was increased beyond the level required for free hover. This produced an upward force greater than that required simply to support the drone, causing the upper tether to become taut and placing the tether network under tension.

Once sufficient tension had been established, the controller input was left essentially unchanged. The drone was not continuously positioned above the central junction and GPS position hold was not used as the mechanism for maintaining the junction.

The drone was observed to move or loiter around the region above the junction. Despite this movement, the central junction remained constrained by the three ground strings and the tension in the upper string.

The experiment therefore relied primarily on mechanical tension rather than continuous positional corrections from the flight controller.

Two conditions were observed to be important:

1. All four strings had to remain taut.
2. The drone needed sufficient angular control to maintain the upper tether under tension.

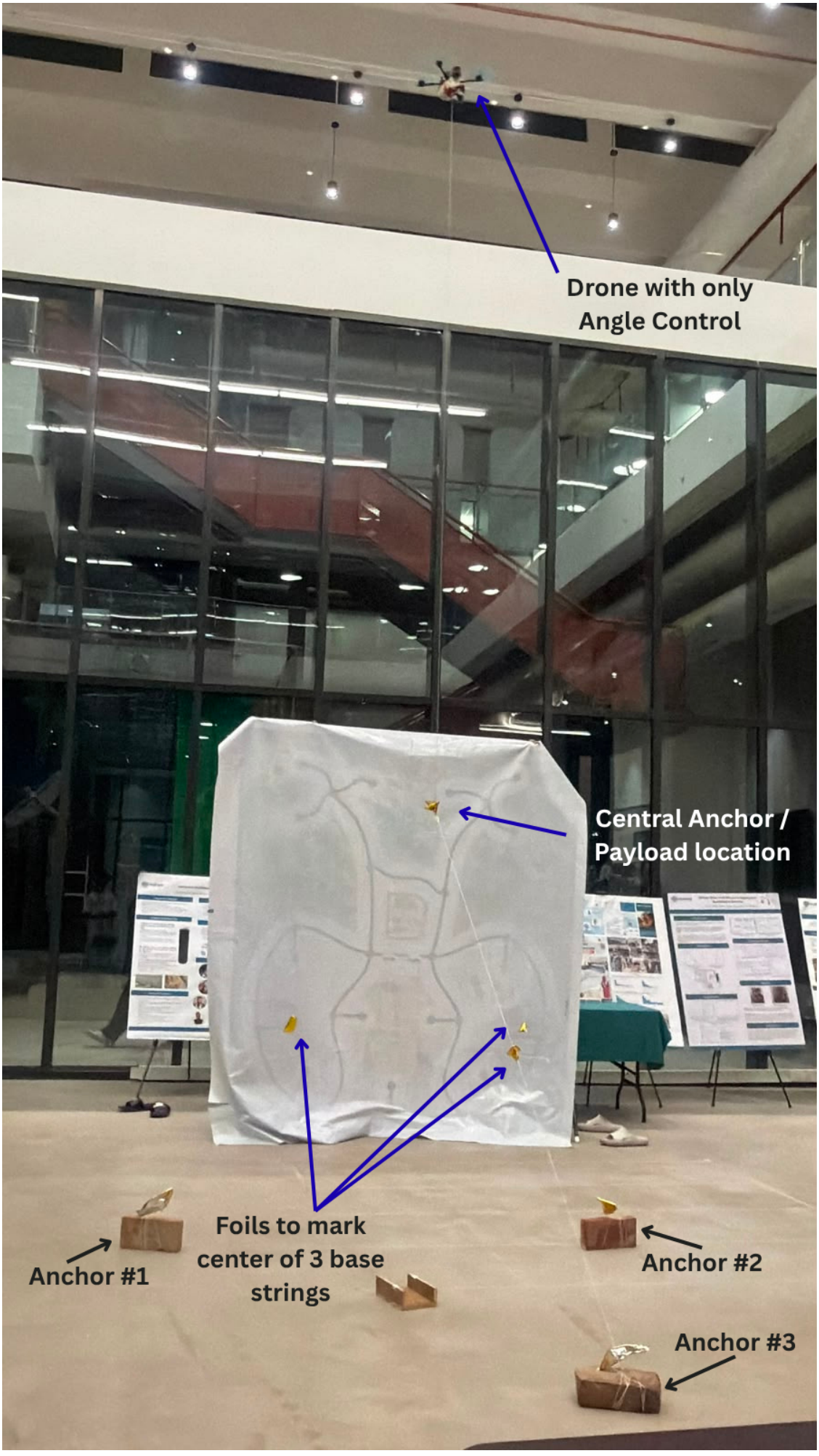


Figure 3: Drone during tether testing with the upper tether under tension.

Observations and Main Finding

The central observation was that the central junction remained spatially constrained even though the drone itself was not maintained at a fixed position.

The drone was allowed to move within the region above the junction while remaining in Angle mode. The increased throttle maintained tension in the upper tether, while the three ground strings constrained the junction from below.

The main finding is therefore:

> *GPS-based positional control of the aerial platform was not necessary for maintaining the central junction's position. Instead, the critical requirement was sufficient angular control of the aerial platform to maintain the orientation and tension of the upper tether.*

This demonstrates an important distinction between **drone localization** and **payload localization**. The drone could move while the mechanically constrained junction remained approximately fixed.

For a static configuration, the junction can be represented by the force balance

$$\mathbf{T}_1 + \mathbf{T}_2 + \mathbf{T}_3 + \mathbf{T}_4 + \mathbf{F}_{\text{payload}} = \mathbf{0}, \tag{1}$$

where the first three terms represent the ground tether tensions and $\mathbf{T}_4$ represents the aerial tether tension.

The exact junction position depends on the anchor geometry, tether lengths, tension, and payload forces. However, the experiment indicates that precise GPS positioning of the aerial platform is not inherently required for the junction to remain constrained.

Discussion

The result changes the requirements for the eventual aerial platform. A conventional drone would normally need to continuously estimate and control its three-dimensional position. In this configuration, the tether system provides much of the mechanical constraint.

The aerial platform primarily needs to provide sufficient force to keep the upper tether taut and sufficient angular control to prevent the tether from becoming slack.

This is particularly important for long-duration operation. A multirotor must continuously generate thrust to remain airborne, making it inefficient for persistent deployment.

The proposed next stage is therefore a **blimp-kite hybrid**. Buoyancy and aerodynamic lift would provide most of the supporting force, while active control would be used primarily for orientation and tension management.



Figure 4: Outdoor test in the night

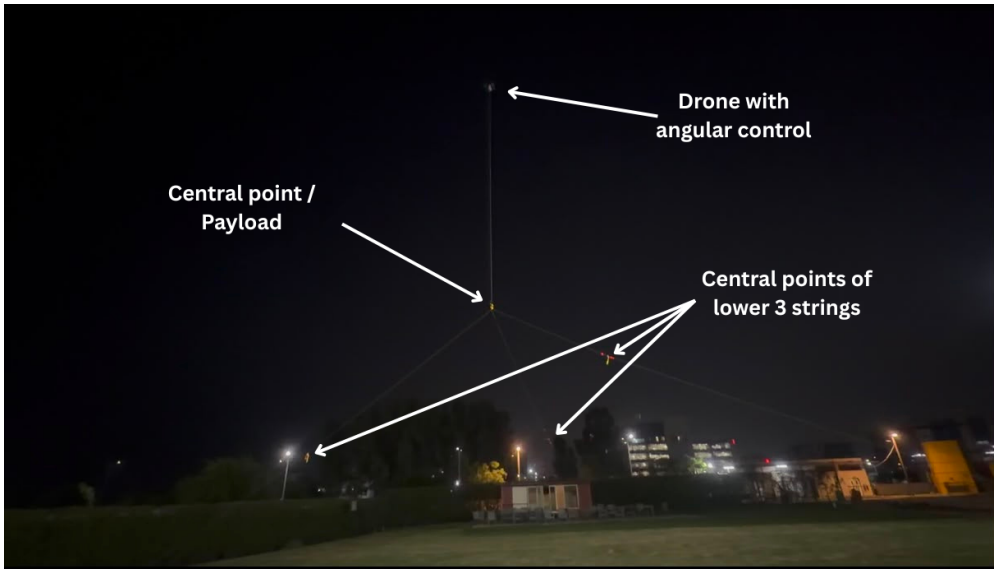


Figure 5: Outdoor test in the night

Energy Efficiency and Endurance

Energy efficiency is a major objective of the proposed system. The final platform is intended to maintain a 1 kg payload at a fixed elevated location for as long as possible, rather than perform conventional flight.

The quadrotor used in the initial experiment continuously consumes power to generate lift, making it unsuitable for very long-duration operation. The proposed blimp-kite platform will instead use buoyancy and aerodynamic lift to provide most of the required supporting force, with active power primarily used for angular control and stabilization.

Future endurance testing will investigate continuous power consumption, operating time, payload mass, wind conditions, and the amount of active control required to maintain tether tension. Since precise positional control was not found to be necessary during the initial experiment, minimizing active control effort is expected to further improve endurance.

The ultimate objective is to maintain a spatially constrained 1 kg payload for extended periods while minimizing continuous energy consumption.

Future Work

The next stage of testing will focus on moving from the initial proof-of-concept to more realistic outdoor operation. Testing will be conducted at greater altitudes and in a wider range of environmental conditions, particularly different wind speeds and directions.

The system will initially be tested with a 1 kg payload, with higher payloads investigated subsequently. Power tethering will also be explored to determine whether continuous ground-based power can be used to extend operating duration without significantly affecting the tether dynamics.

Further testing will investigate different tether lengths and anchor geometries, the effect of wind on the central junction, and the minimum angular control required to maintain tension. More extensive outdoor testing will ultimately be used to evaluate the stability, payload capacity, energy consumption, and endurance of the system under realistic operating conditions.

Conclusion

The experiment demonstrated that the central junction can remain spatially constrained without GPS-based positional control of the aerial platform. As long as the tether system remains taut, the primary control requirement is sufficient **angular control of the aerial platform to maintain tether tension**. This finding supports replacing the quadrotor with a more energy-efficient blimp-kite platform for long-duration 1 kg payload deployment.